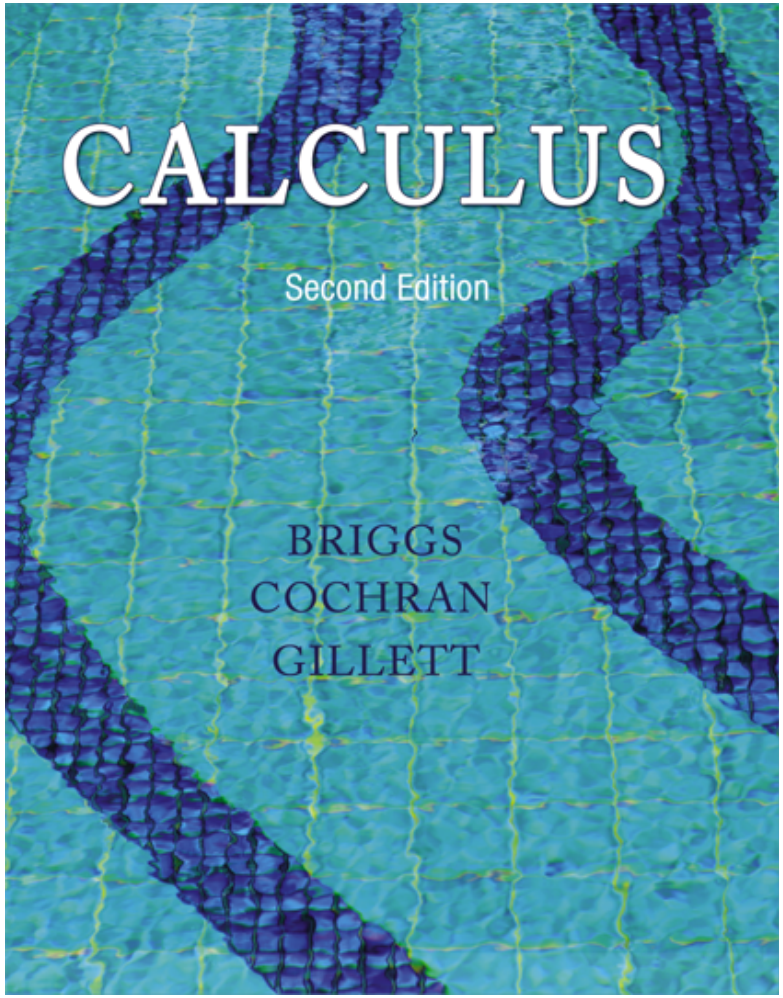
Chapter 2

Limits



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2.7

Precise Definitions of Limits

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Figure 2.54

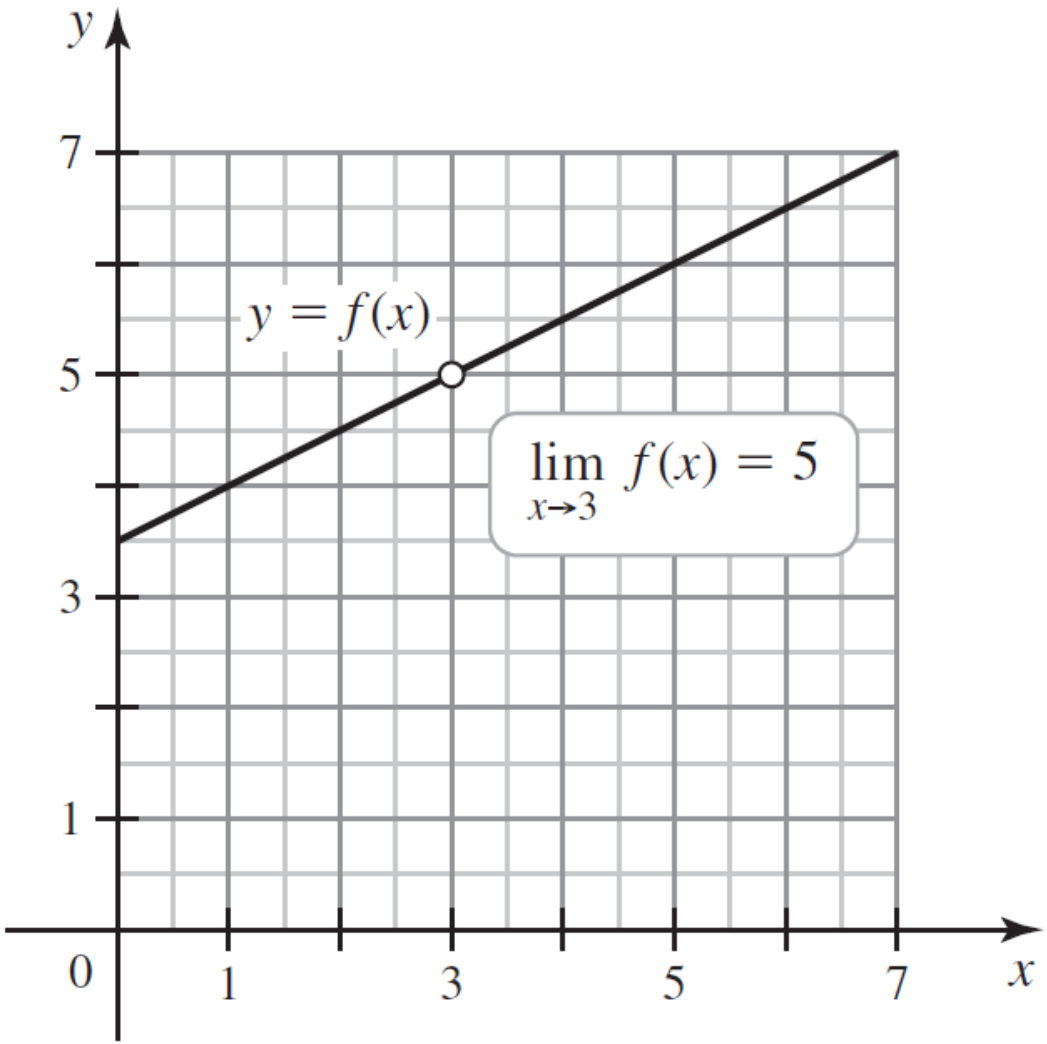

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Figure 2.55 (a)

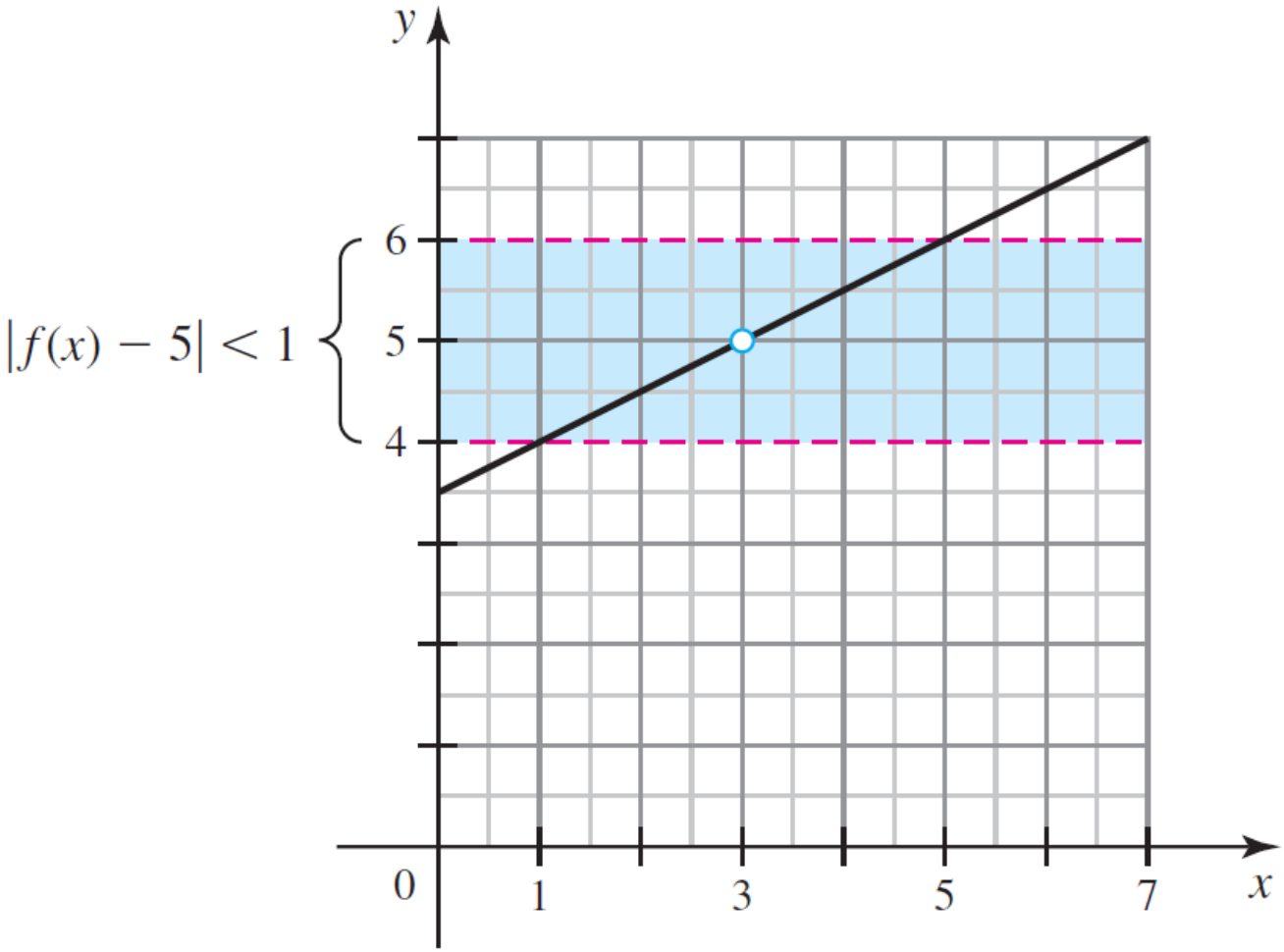

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Figure 2.55 (b)

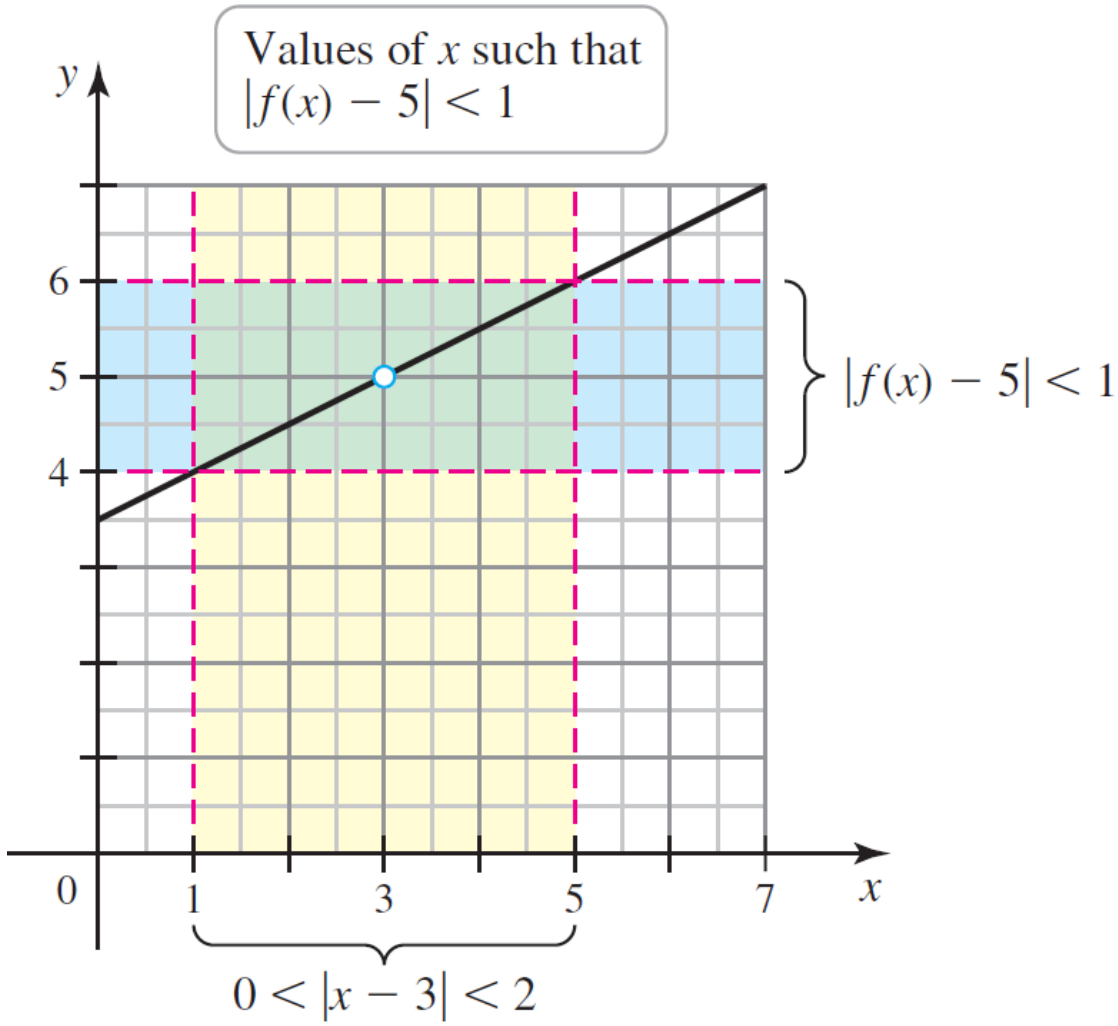

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Figure 2.56 (a)

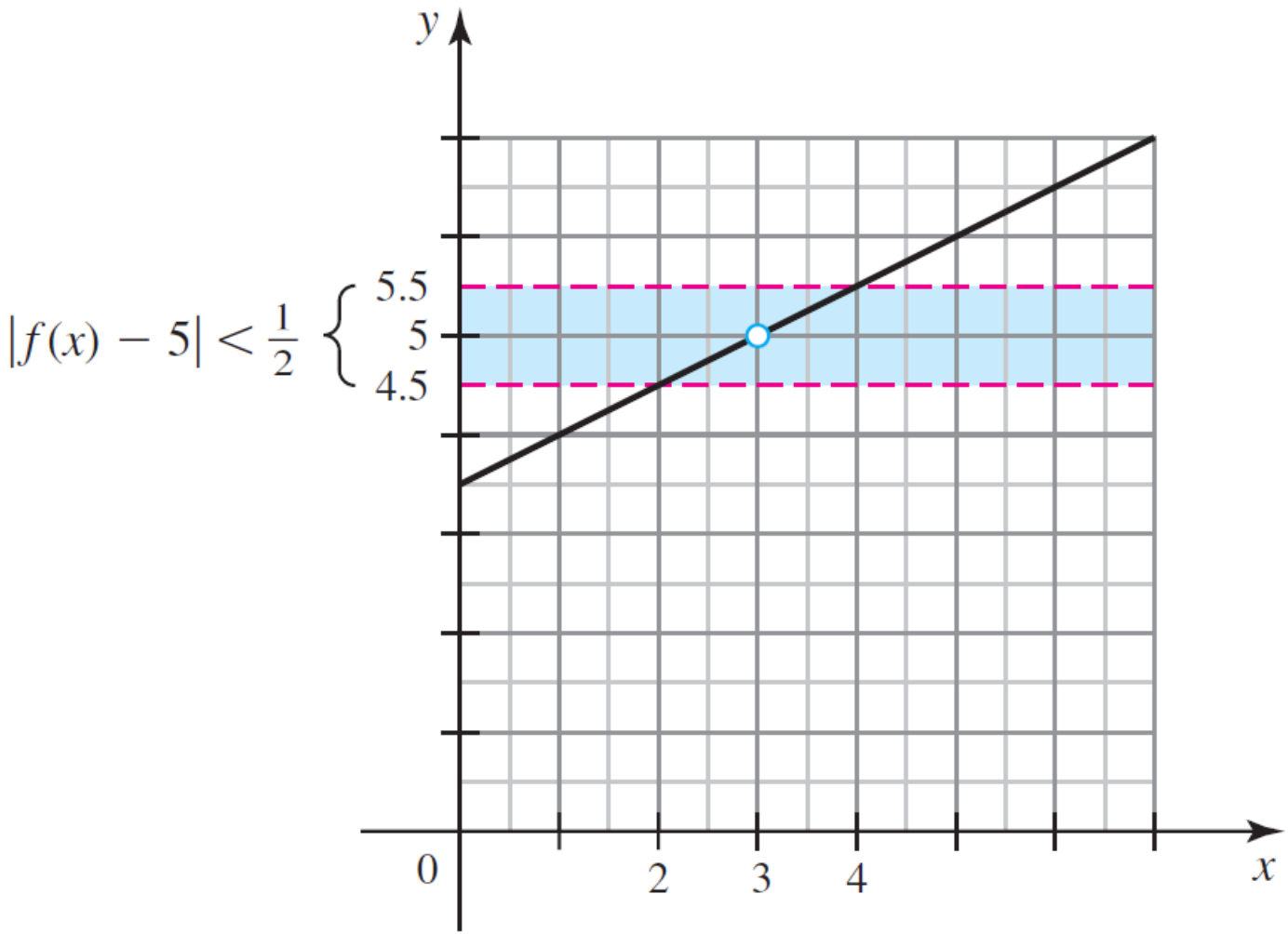

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Figure 2.56 (b)

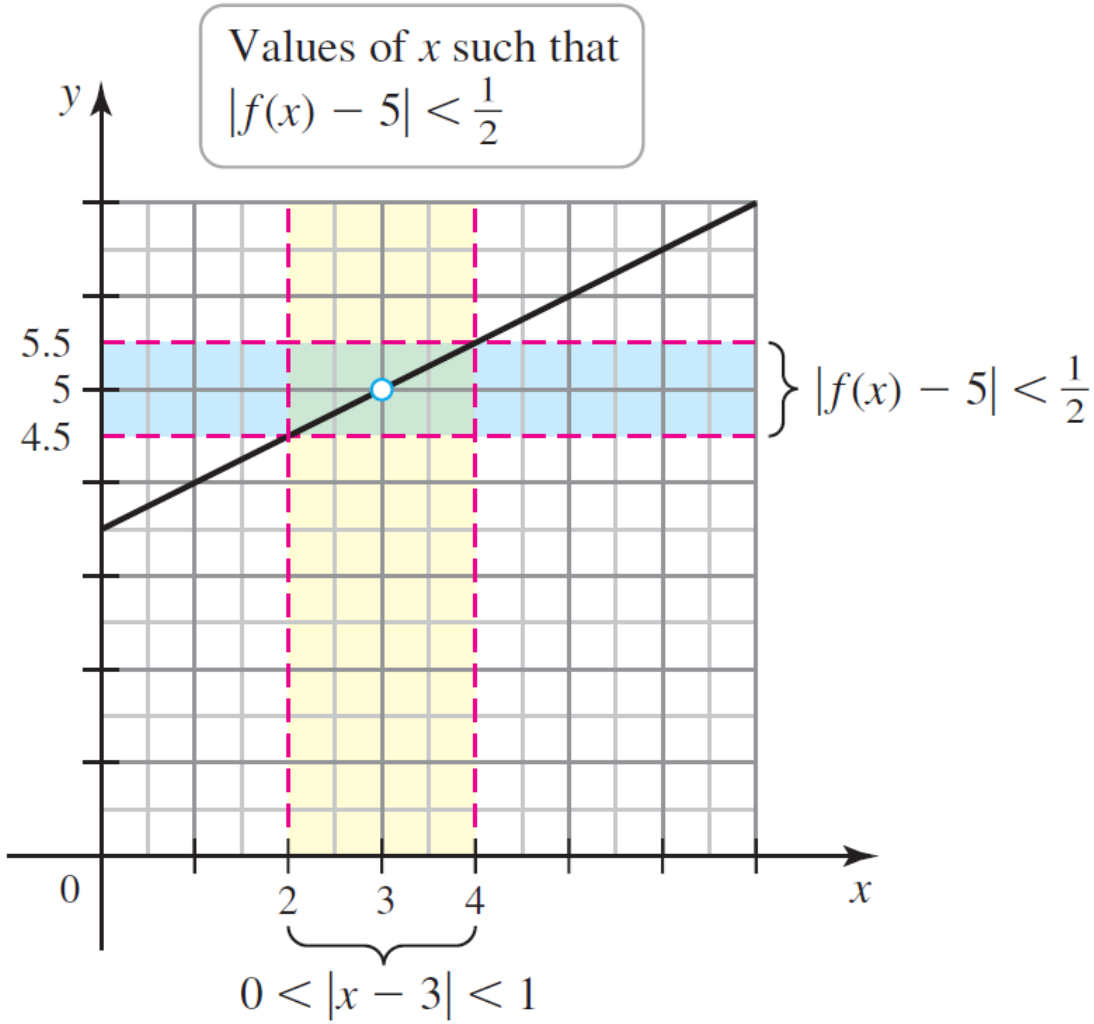

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Figure 2.57

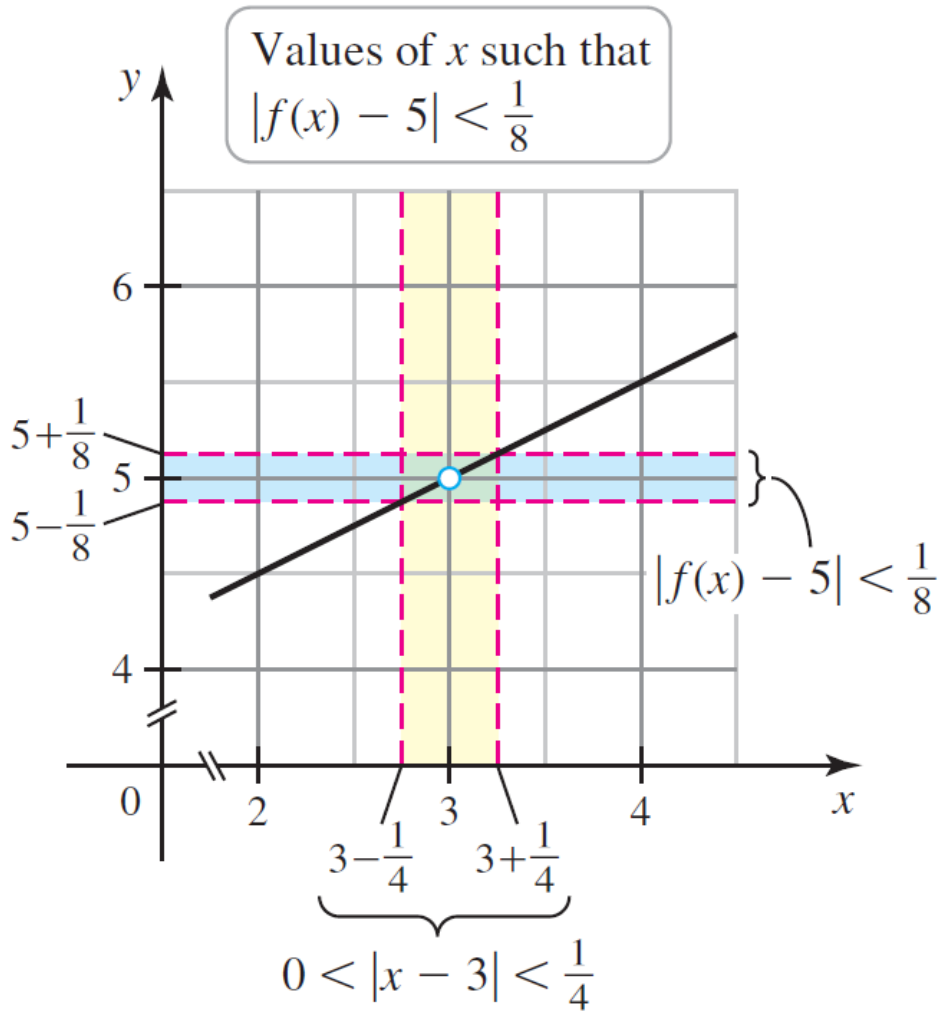

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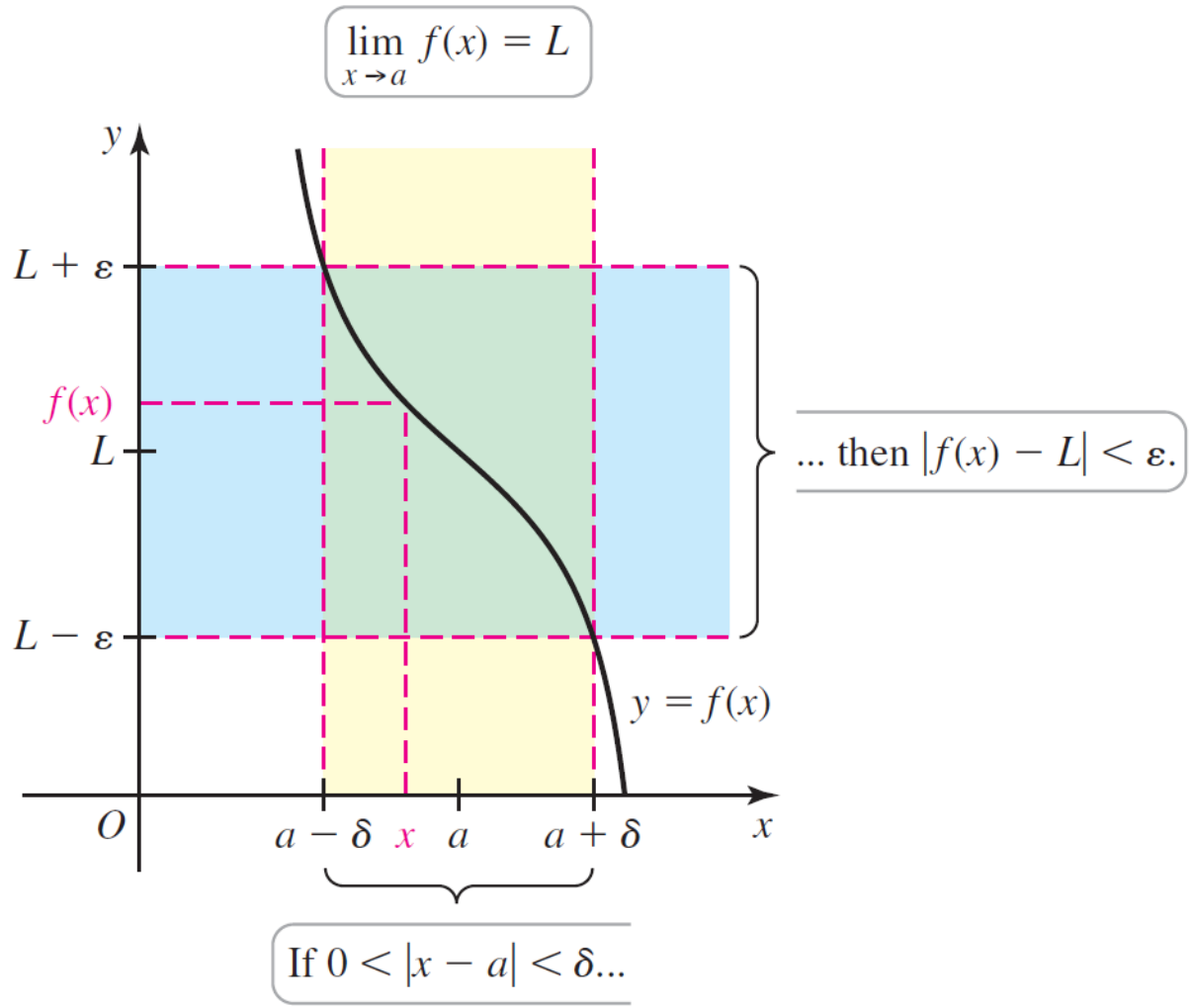
Figure 2.58



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DEFINITION Limit of a Function

Assume that $f(x)$ exists for all x in some open interval containing a, except possibly at a. We say that the **limit of $f(x)$ as x approaches a is L**, written

$$\lim_{x \to a} f(x) = L,$$

if for *any* number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta.$$

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Figure 2.59

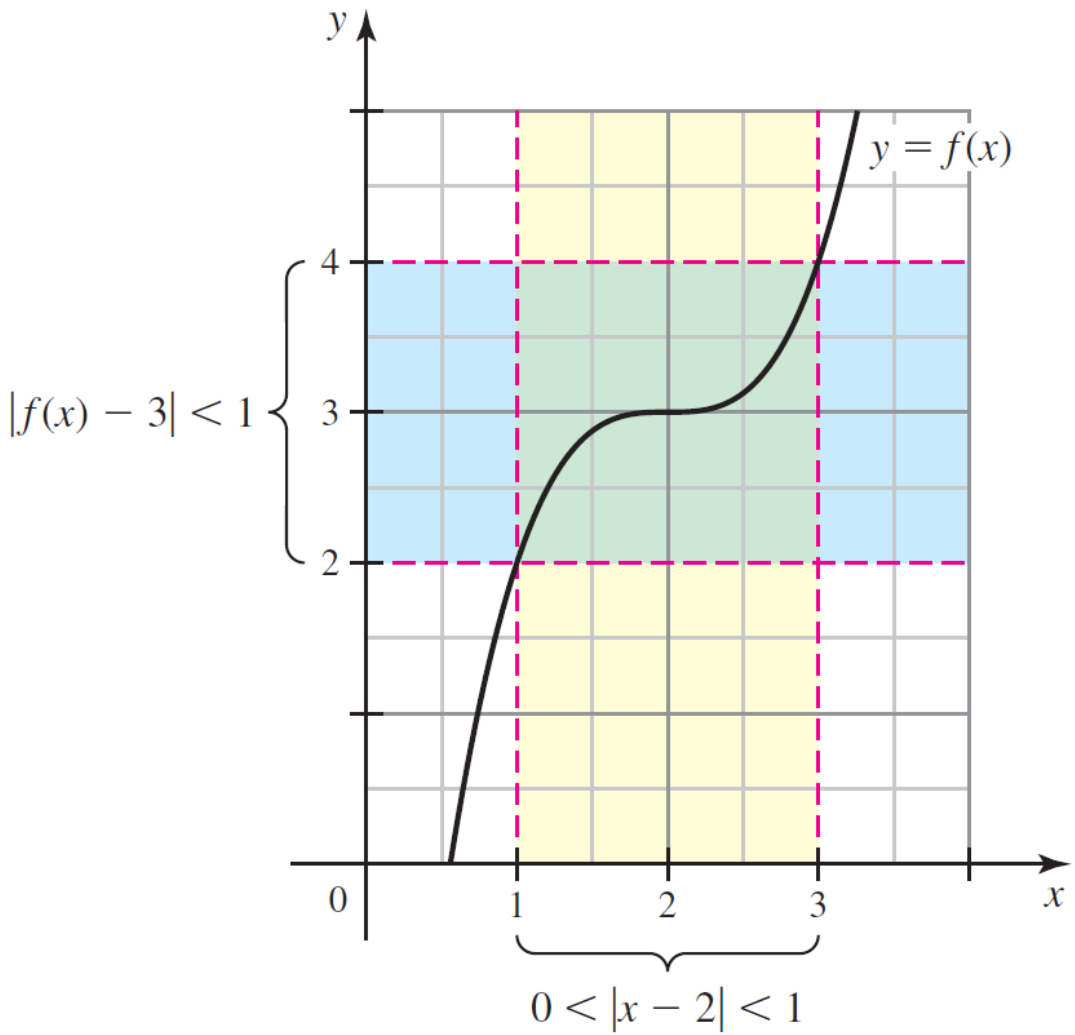

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Figure 2.60

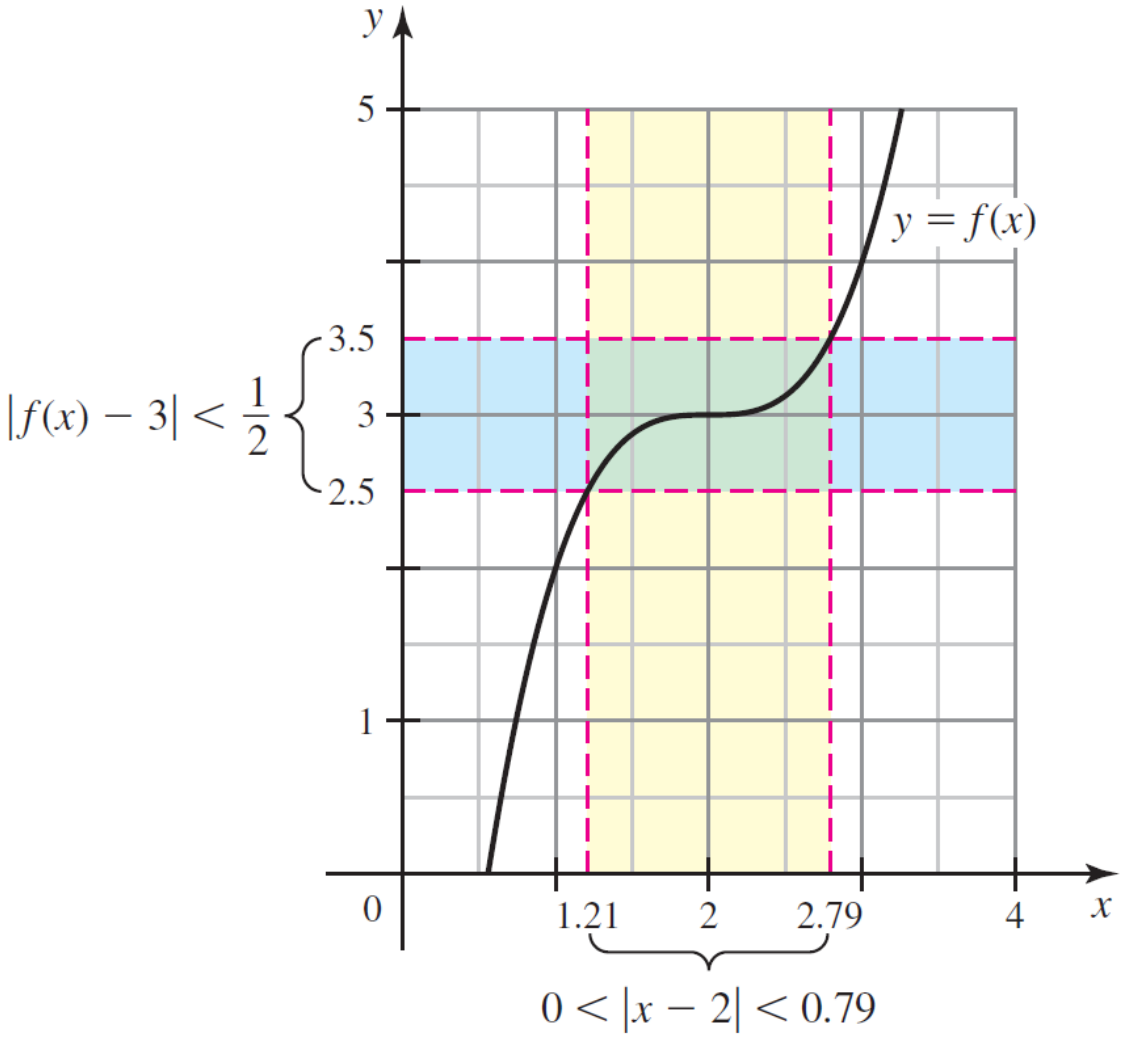

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Figure 2.61

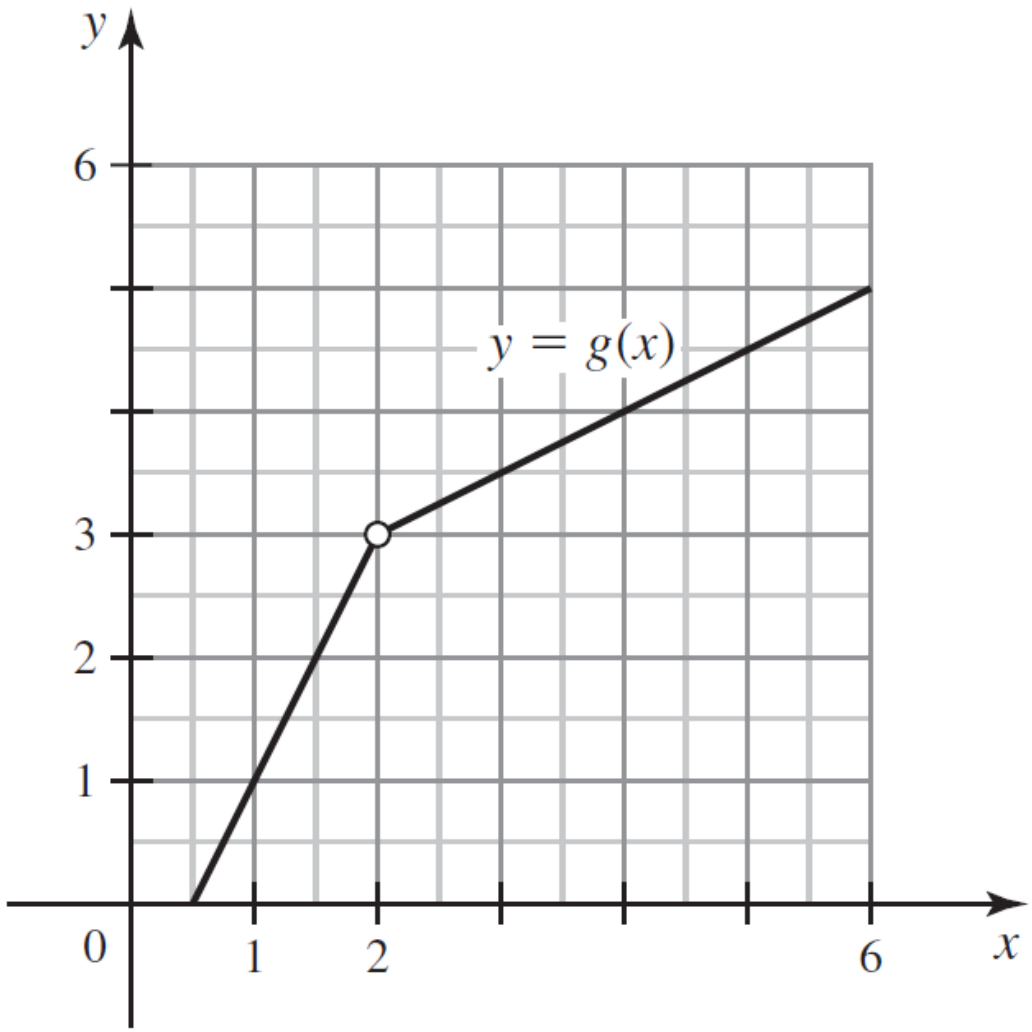

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Figure 2.62 (a)

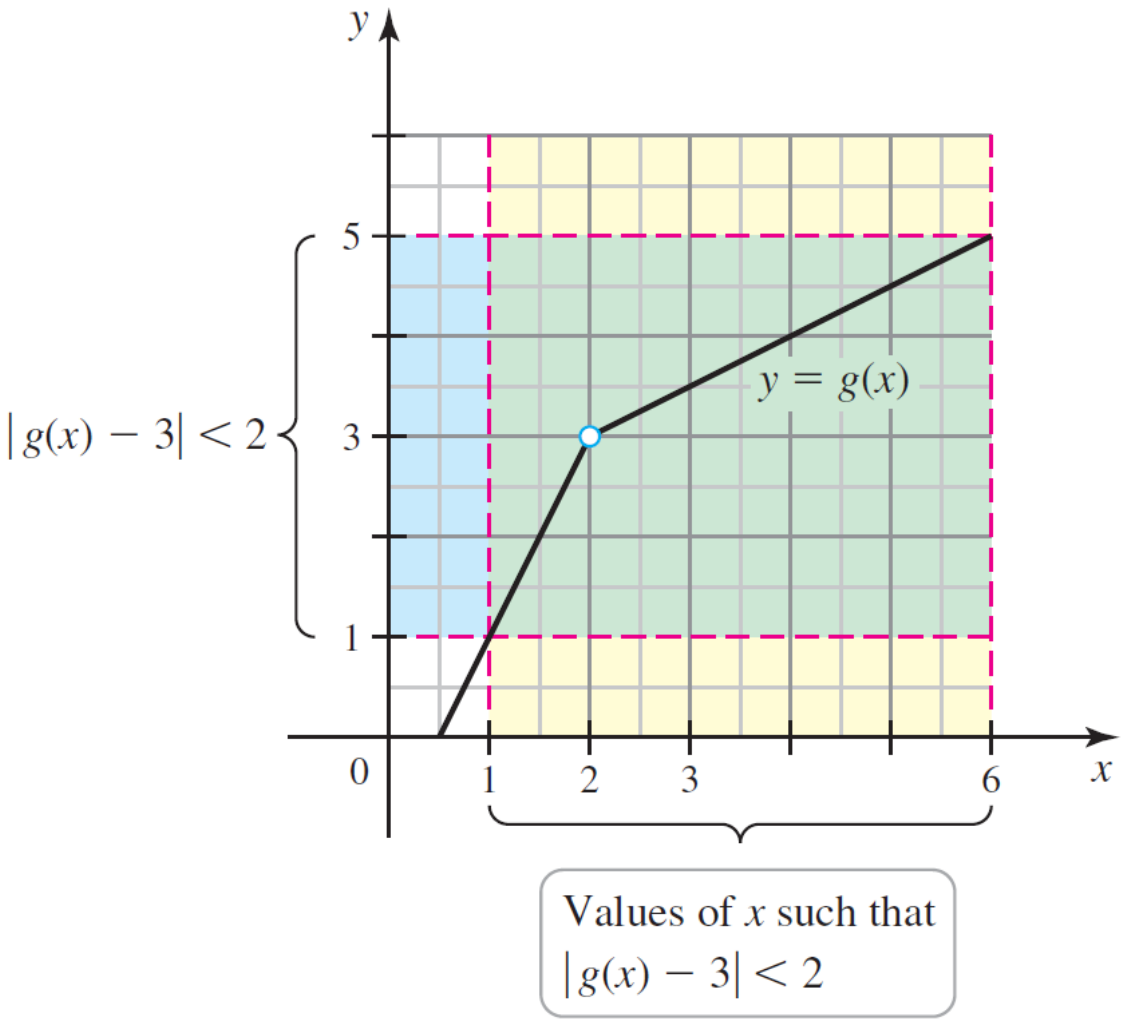

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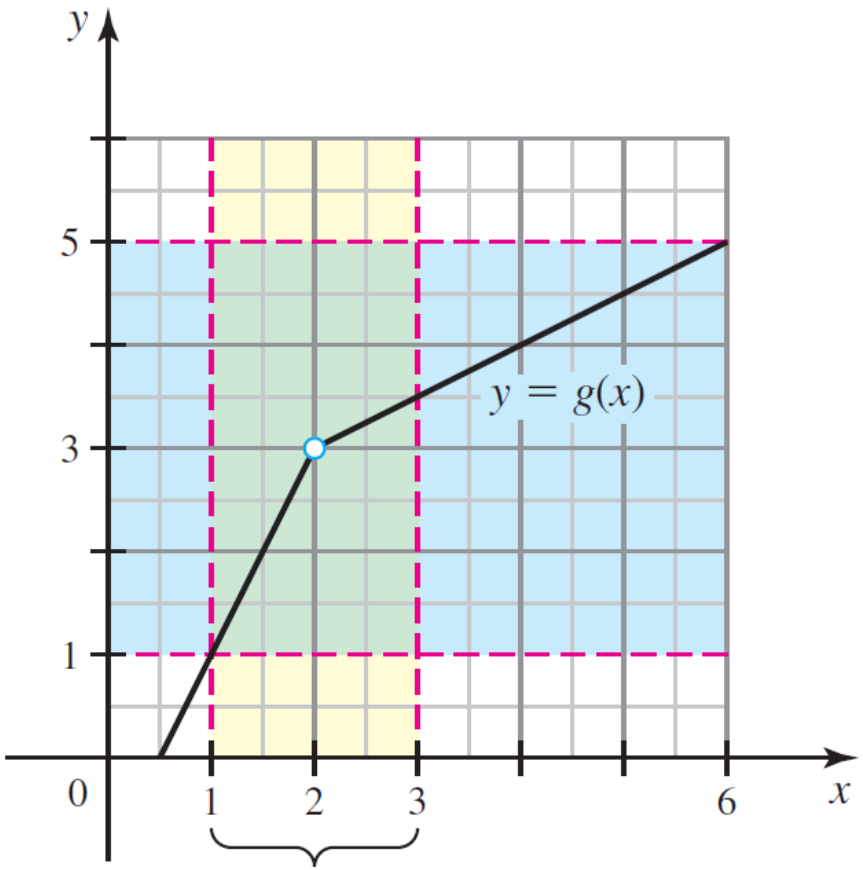
Figure 2.62 (b)



Symmetric interval $0 < |x - 2| < 1$
that guarantees $|g(x) - 3| < 2$

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Figure 2.63 (a)

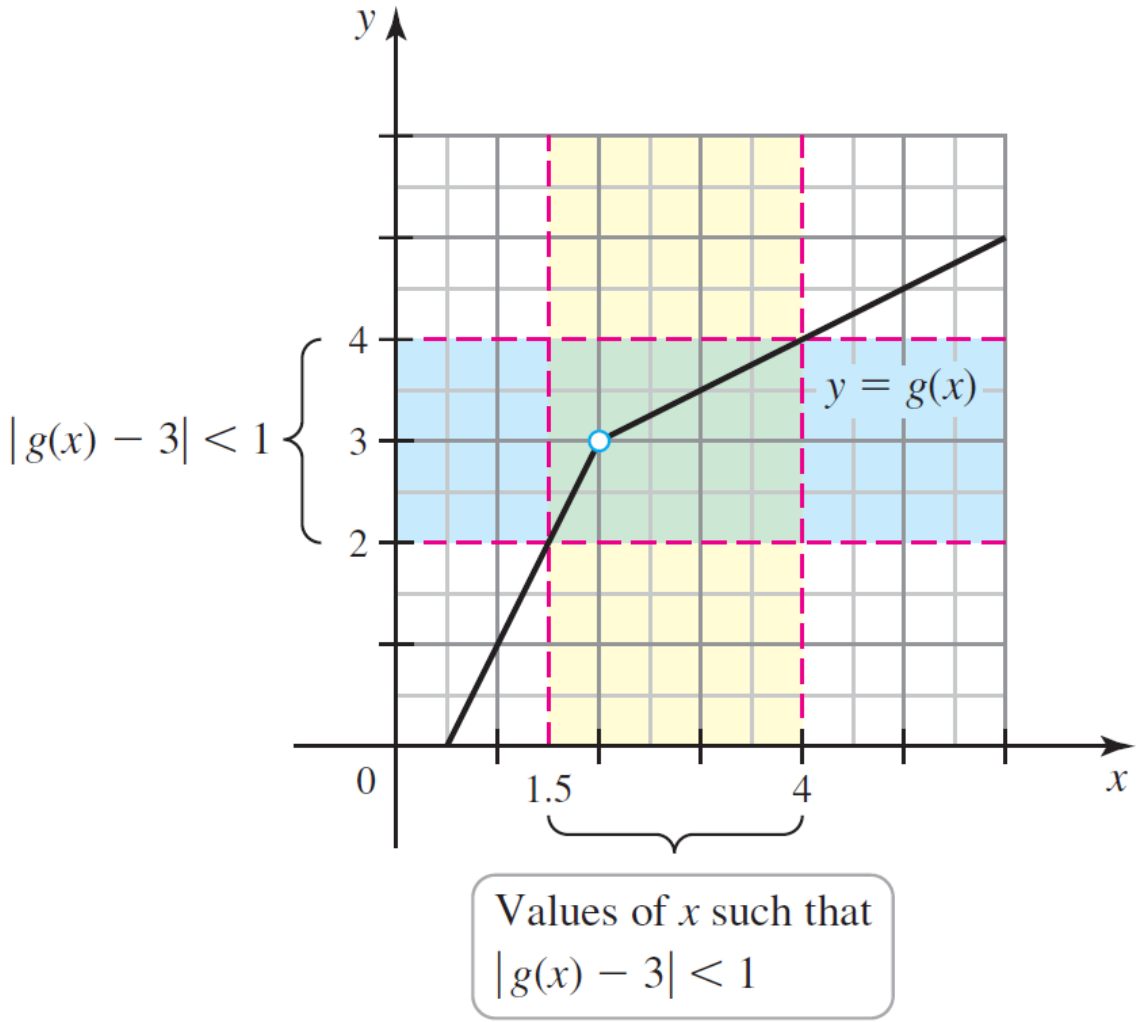

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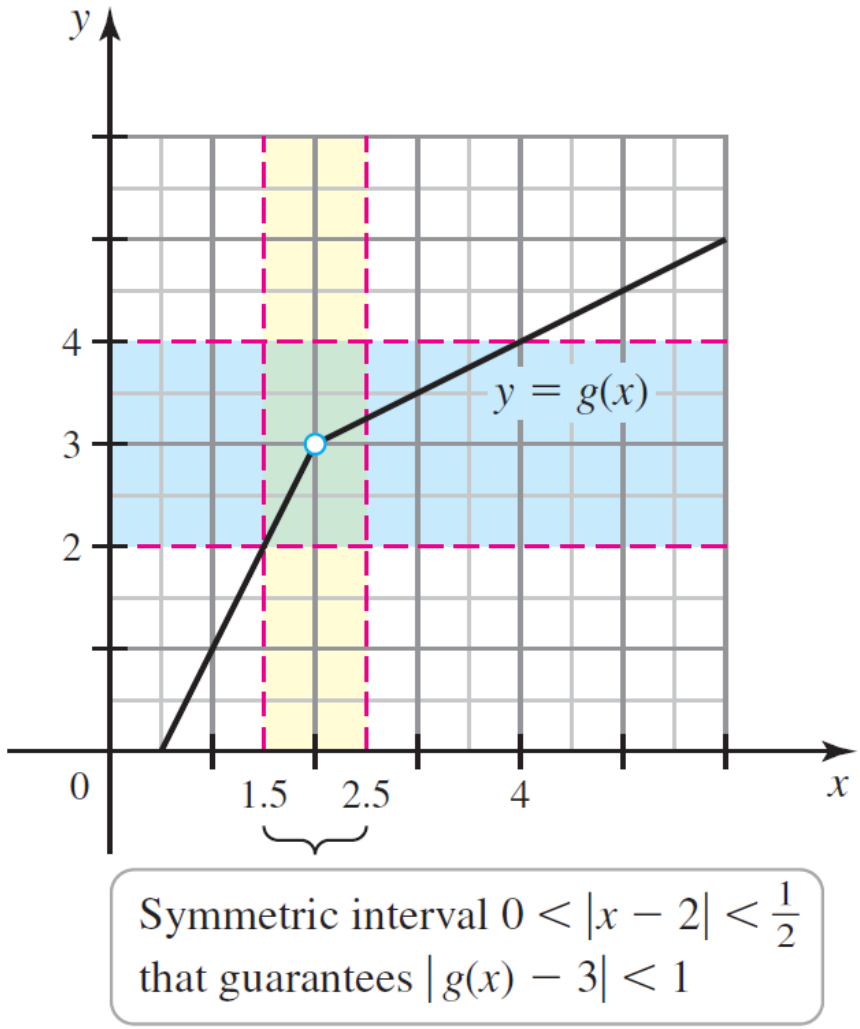
Figure 2.63 (b)



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Steps for proving that $\lim_{x \to a} f(x) = L$

1. **Find δ.** Let ε be an arbitrary positive number. Use the inequality $|f(x) - L| < \varepsilon$ to find a condition of the form $|x - a| < \delta$, where δ depends only on the value of ε.

2. **Write a proof.** For any $\varepsilon > 0$, assume $0 < |x - a| < \delta$ and use the relationship between ε and δ found in Step 1 to prove that $|f(x) - L| < \varepsilon$.

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DEFINITION Two-Sided Infinite Limit

The **infinite limit** $\lim_{x \to a} f(x) = \infty$ means that for any positive number N there exists a corresponding $\delta > 0$ such that

$$f(x) > N \quad \text{whenever} \quad 0 < |x - a| < \delta.$$

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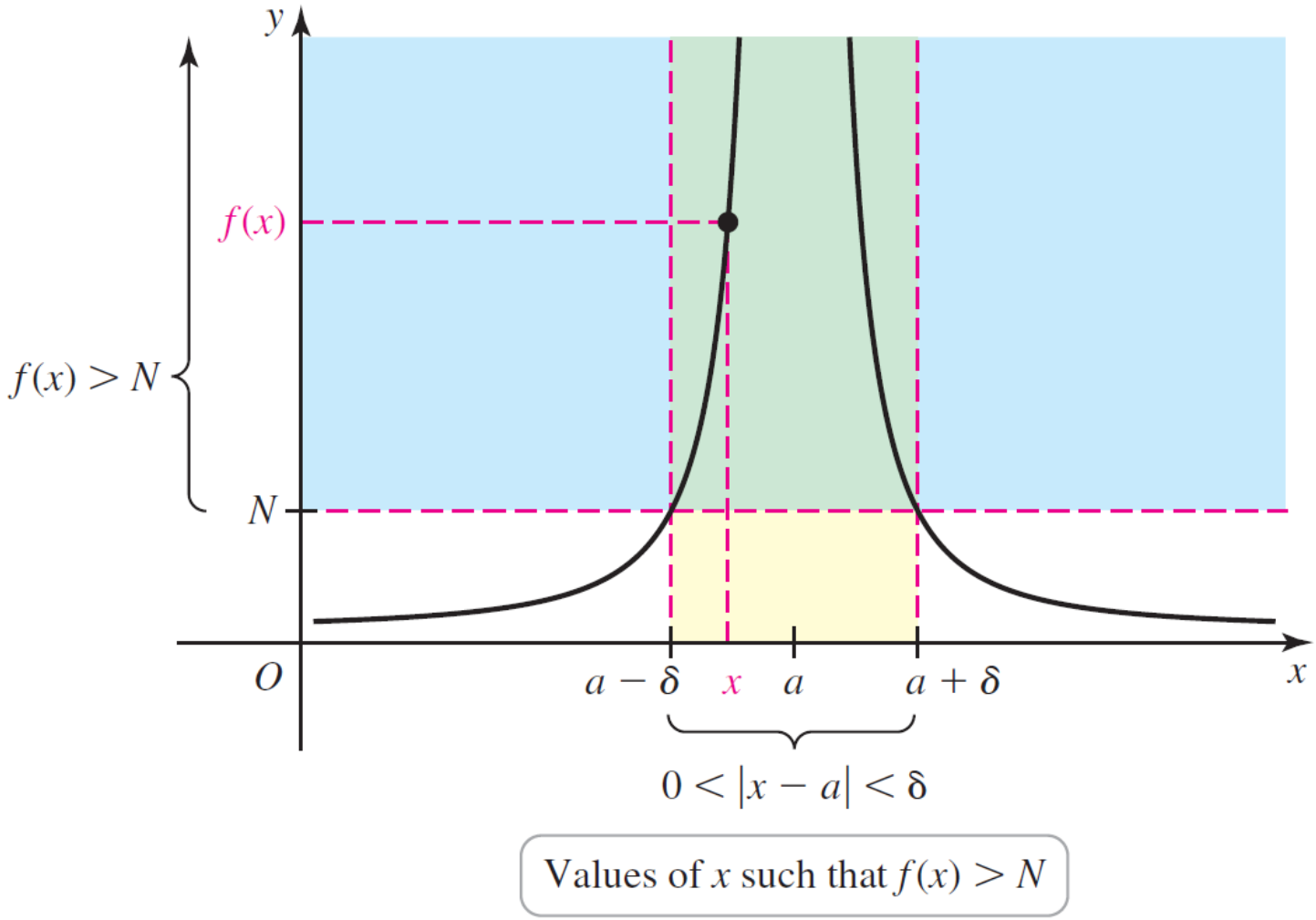
Figure 2.64



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Steps for proving that $\lim_{x \to a} f(x) = \infty$

1. **Find δ.** Let N be an arbitrary positive number. Use the statement $f(x) > N$ to find an inequality of the form $|x - a| < \delta$, where δ depends only on N.

2. **Write a proof.** For any $N > 0$, assume $0 < |x - a| < \delta$ and use the relationship between N and δ found in Step 1 to prove that $f(x) > N$.

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